

NICOLAS VIULET

viuletn@mcmaster.ca — 248-678-8752 — Hamilton, ON

EDUCATION

MSc., Master of Science

McMaster University

Sep. 2025 — Present

Hamilton, ON

- **Program:** Neuroscience
- **Thesis:** “Investigating Intracortical Myelin as a Biomarker for Depression Severity in Mood Disorders”
- **Thesis Supervisor:** Dr. Nicholas Bock
- **Selected Coursework:** Statistical Methods in Neuroscience

HBSc., Honours Bachelor of Science

University of Toronto

Sep. 2021 — Jun. 2025

Toronto, ON

- **Program:** Neuroscience Specialist Major | Buddhism, Psychology, and Mental Health Minor
- **Thesis:** “Acute Functional Brain Changes Associated with Psilocybin in Depression: A Protocol for a Systematic Review and Meta-Analysis”
- **Thesis Supervisor:** Dr. Sean Nestor

HONOURS AND AWARDS

- Dean’s List Scholar in the Faculty of Arts & Science | University of Toronto (2023, 2025)
- James Morrow Scholarship | University of Toronto (2023)
- The Dr. Lorus J. Milne and Dr. Margery J. Milne Research Award | University of Toronto (2025)
- Lincoln Hutton Essay Prize: Sciences | University of Toronto (2025)

PUBLICATIONS & ABSTRACTS

ABSTRACTS (REFEREED)

1. Poulin, J., **Viulet, N.**, Liu, A., MacIntosh, B. J., & Nestor, S. M. (2026, May). Acute Functional Brain Changes Associated With Psilocybin in Healthy Adults and Depression: A Systematic Review and Meta-Analysis. *Biological Psychiatry*, 99(10), S134. <https://doi.org/10.1016/j.biopsych.2026.03.229>

CONFERENCE PRESENTATIONS (REFEREED)

1. **Viulet, N.**, Naseem, S., Wong, A., Hassel, S., Khoshroo, S., Rowley, C., Churchill, N., Dunlop, K., Minuzzi, L., Frey, B., & Bock, N. (2026, August). Investigating Intracortical Myelin as a Biomarker for Depression Severity in Mood Disorders. ISMRM (International Society for Magnetic Resonance in Medicine) Workshop on Advancing Translational MR Neuroimaging: Towards Imaging Biomarkers in Psychiatry and Neurodegeneration. Saint-Sulpice, Switzerland. (Oral and Poster Presentation).
2. **Viulet, N.**, Naseem, S., Wong, A., Hassel, S., Khoshroo, S., Rowley, C., Churchill, N., Dunlop, K., Minuzzi, L., Frey, B., & Bock, N. (2026, May). Establishing Intracortical Myelin as a Biomarker for Depression Severity in a Mood Disorder Population. Southern Ontario Neuroscience Association (SONA) Annual Meeting. Hamilton, ON. (Poster Presentation).

CONFERENCE PRESENTATIONS (INVITED)

1. **Viulet, N.**, Naseem, S., Wong, A., Hassel, S., Khoshroo, S., Rowley, C., Churchill, N., Dunlop, K., Minuzzi, L., Frey, B., & Bock, N. (2026, April). Investigating the Relationship Between Intracortical Myelin and Depression Severity in Mood Disorders. Psychology Neuroscience and Behavior & Neuroscience Graduate Research Day. Hamilton, ON. (Poster Presentation).
2. Poulin, J., **Viulet, N.**, Liu, A., MacIntosh, B., & Nestor, S. (2025, July). Acute Functional Brain Changes Associated with Psilocybin in Healthy Adults and Depression: A systematic review and meta-analysis. SRI (Sunnybrook Research Institute) Research Day. Toronto, ON. (Poster Presentation).

3. Poulin, J., **Viulet, N.**, Liu, A., MacIntosh, B., & Nestor, S. (2025, June). Acute Functional Brain Changes Associated with Psilocybin in Healthy Adults and Depression: A systematic review and meta-analysis. University of Toronto Department of Psychiatry Research Day. Toronto, ON. (Poster Presentation).
4. **Viulet, N.** (2025, March). Magic Mushrooms: A New Frontier in Mental Health. Victoria College Research Day. Toronto, ON. (Poster Presentation).

OTHER PUBLICATIONS (NON-REFEREED WRITING)

1. **Viulet, N.** (2026, May). OCD and Predictive Processing: Remediating Reduced Self-Confidence and Negative Affect with Meditation. UPAYA Buddhism and Psychology Student Union Journal, (11), 43-52. <https://bpsublog.wordpress.com/upaya-journal/>
2. **Viulet, N.** (2023, December). Effects of Aerobic Exercise on the Brain: Benefits of Aerobic Exercise for Parkinson's Disease Patients. Project Wellness Student Journal, (4), 36-44. <https://heyzine.com/flip-book/6704ae2bbe.html#page/37>

RESEARCH EXPERIENCE

MSc Thesis Student

Sep. 2025 — Present

Neuroscience Program, McMaster University

Hamilton, ON

- Developed an M.Sc. thesis project evaluating intracortical myelin (ICM), across 360 regions of interest (ROIs), as a biomarker for depression severity in mood disorders using the quantitative magnetic resonance imaging (qMRI) parameter R_1 and the Montgomery-Åsberg Depression Rating Scale (MADRS) to measure ICM and depression severity respectively.
- Processed multi-site qMRI data for 100+ participants using a pipeline that used FreeSurfer, FSL, MATLAB, and Connectome Workbench on SLURM-managed computing clusters; this data was then analyzed in R using a multiple linear regression approach.

Undergraduate Thesis Student (HMB496Y1)

Jul. 2024 — Apr. 2025

EMBRACE Project, Sunnybrook Health Sciences Centre

Toronto, ON

- Conducted a PRISMA-compliant and PROSPERO registered systematic literature search of 1577 research papers to evaluate pre- to post-psilocybin functional connectivity shifts in mood and emotion networks in healthy versus depressed adults.
- Consolidated region- and network-level therapeutic targets (e.g., sgACC, dlPFC, salience network, DMN) to support a-priori regions of interest for the Engaging Mood Brain Circuits with Psilocybin (EMBRACE) clinical trial.

Undergraduate Research Assistant (Work-Study & Volunteer)

Aug. 2023 — Aug. 2025

The BOLD Lab, OISE, University of Toronto

Toronto, ON

- Administered standardized English psychometric evaluations and acquired functional neural activity using fNIRS across ~120 participants for the Reading Development in Resettled Refugees project, securing data via REDCap.
- Maintained and updated lab data by using Python scripts to automate data organization, performed quality control cleaning on psychometric data, and onboarded/trained new research assistants on fNIRS and behavioral administration protocols.

TEACHING AND MENTORSHIP EXPERIENCES

Science ISC1 Graduate Mentor

Aug. 2026 — Present

*Faculty of Science, McMaster University**Hamilton, ON*

- Mentoring first-year science students during their transition to university, facilitating weekly engagement sessions, and providing guidance on academic navigation and university resources.

Teaching Assistant (PNB 2XB3: Neuroanatomy & Neurophysiology)

Sep. — Dec. 2025 & 2026

*Department of Psychology, Neuroscience & Behaviour, McMaster University**Hamilton, ON*

- Evaluated course assignments and provided detailed academic feedback under the supervision of Dr. Daniel Goldreich.

Undergraduate Academic Mentor

Jan. 2024 — Apr. 2025

*Ask An Undergrad (AAU), University of Toronto Chapter**Toronto, ON*

- Conducted structured mentorship sessions and coffee chats for incoming first-year students to support their transition into post-secondary education and long-term academic planning.

Undergraduate Academic Mentor

Nov. 2023 — Apr. 2024

*Neurological and Neurodegenerative Disease Student Association (NANSA)**Toronto, ON*

- Mentored first- and second-year Neuroscience students by guiding mentees on how to secure undergraduate research positions, apply for research funding, select meaningful courses, and explore post-graduation career pathways.

WORK EXPERIENCE

First Aid & CPR Instructor

Jun. 2026 — Present

*St. John Ambulance**Hamilton, ON*

- Instructed and evaluated candidates in Standard and Emergency First Aid along with CPR Level A, Level C, and AED operation through demonstrating lifesaving emergency response protocols, assessing candidate practical skills, and ensuring adherence to first aid training standards.

MEMBERSHIPS

Trainee Member

Sep. 2025 — Present

*Canadian Biomarker Integration Network in Depression (CAN-BIND)**Canada (National Organization)*

- Participated in multi-site translational research initiatives, network-wide scientific meetings, and interdisciplinary training focused on biomarker discovery in major depressive disorder (MDD).

Affiliated Researcher

Jan. 2025 — Present

*Embodiment, Attention, Sensation, Engagement (EASE) Lab**Toronto, ON*

- Engaged in monthly interdisciplinary meetings bringing together neuroscientists, clinicians, and contemplative scholars to collaborate on mind-body research.

SERVICE

Farm Volunteer

May. 2026 — Present

*Manorun Organic Farm**Hamilton, ON*

- Planted, weeded, and harvested produce throughout the summer growing season, gaining firsthand appreciation for the sustained physical effort, patience, and attention to detail required to practice sustainable agriculture.

Event Volunteer Organizer*SONA 2026 Annual Meeting, McMaster University*

May. 2026

Hamilton, ON

- Coordinated event logistics, schedule structure, and venue arrangements for the regional neuroscience conference hosted at McMaster University.

Volunteer Judge*NeuroXChange Undergraduate Research Conference, McMaster University*

Apr. 2026

Hamilton, ON

- Evaluated 5 undergraduate student poster presentations ensuring that students received feedback that served to motivate further exploration of their research interests.

Events Team Executive Member*Student of the MiNDS Association (SOMA), McMaster University*

Sep. 2025 — Aug. 2026

Hamilton, ON

- Oversee event planning, venue logistics, and budget allocation for the Neuroscience Graduate Students' Union to foster community engagement among graduate trainees.

Treasurer*Buddhism, Psychology, and Mental Health Student Union (BPSU), University of Toronto*

Apr. 2024 — Aug. 2025

Toronto, ON

- Managed a \$4,000 CAD annual budget, allocated by the Arts and Science Student Union (ASSU), to support the execution of academic seminars, student workshops, and social events.

Certified Nurse Assistant (CNA) Skills Trainer*Michigan Healthcare Academy (MHA)*

Jun. 2022 — Apr. 2023

Mt. Clemens, MI

- Collaborated with senior teachers to teach essential clinical skills to nursing assistant students, through lectures and demonstrations, to prepare them for state certification examinations such including, but not limited to, vital sign acquisition, patient locomotion, perineal care, and specialized care protocols for patients with neurological conditions.

SKILLS AND CERTIFICATIONS

Healthcare

- Oxygen Therapy — *Canadian Red Cross* (ID: 104320537)
- Airway Management — *Canadian Red Cross* (ID: 104320535)
- Standard First Aid CPR/AED Level C — *St. John Ambulance* (ID: 0568830462)

Research & Ethics

- Responsible Conduct of Research (RCR) — *CITI Program* (ID: 68697823)
- Biomedical Research Ethics Tutorial — *CITI Program* (ID: 57746125)
- Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS2) — (ID: 00014199954)
- Health Canada Division 5: Drugs for Clinical Trials Involving Human Subjects — *CITI Program* (ID: 68697824)
- Good Clinical Practice (GCP) — *CITI Program* (ID: 68697822)

Technical Knowledge

- **Neuroimaging Modalities & Analysis:** FreeSurfer, FSL, Connectome Workbench, ITK-SNAP, fNIRS, Structural & Quantitative MRI (R1/T1 mapping, Cortical Thickness), Diffusion Tensor Imaging (DTI), Functional MRI (fMRI)
- **Programming Languages & Data Science:** R, Python, MATLAB, HTML
- **High-Performance Computing & Tools:** Linux/Unix environment, SLURM Workload Manager, REDCap

Languages

- English (Native)
- Romanian (Native)